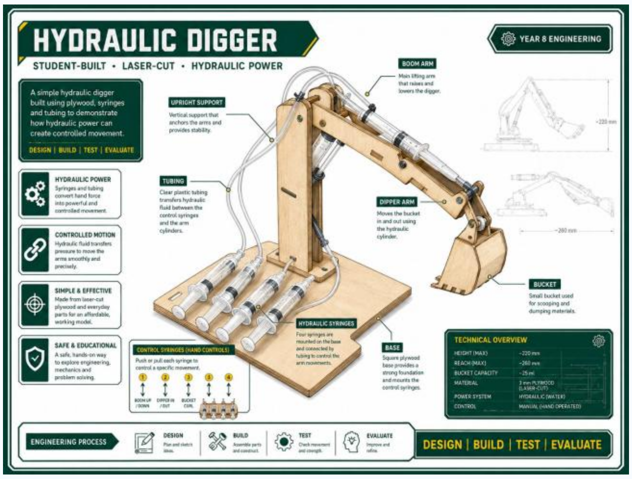
PROJECT BRIEF

A strong answer defines the design challenge, success criteria, and the evidence that will prove the digger works.



1 Design challenge

- Create a tabletop hydraulic digger
- Use boom, dipper and bucket movement
- Transfer force with syringes and tubing

2 Success criteria

- 3 circuits: boom, dipper, bucket
- Smooth, leak-free motion
- Reach, lift and control evidence

3 Evidence to collect

- Sketches and pivot layout
- Syringe sizes and calculations
- Test data, photos and evaluation

OK

BEST PRACTICE STUDENT RESPONSE

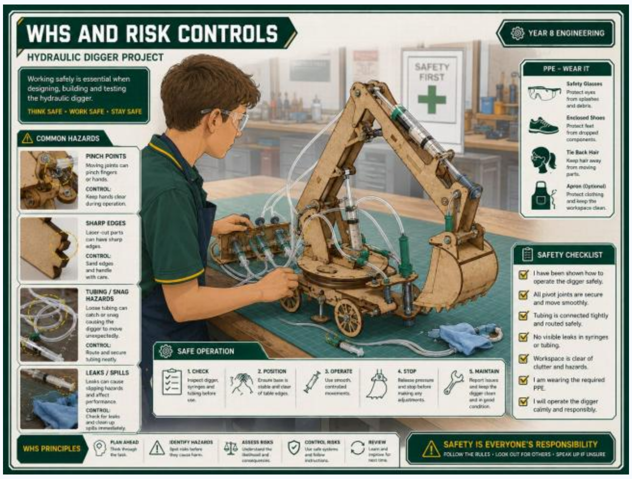
The design challenge was to build a tabletop hydraulic digger with three controlled hydraulic movements. A successful digger needed smooth, leak-free boom, dipper and bucket action. My folio will use sketches, photos, test data and calculations to prove reach, lift and control.

BEFORE YOU TICK DONE

- ☐ Brief stated
- ☐ Criteria named
- ☐ Evidence planned

WHS AND RISK CONTROLS

A strong answer explains the hazards and shows how safe workshop practice controlled the risks.



1 Main hazards

- Drilling and cutting hazards
- Pinch points in linkages
- Water spills near the work area

2 Risk controls

- Wear PPE and follow tool rules
- Clamp work before drilling
- Keep fingers clear of moving arms

3 Safe testing

- Clear test area
- Control water leaks quickly
- Keep hands away during movement

OK

BEST PRACTICE STUDENT RESPONSE

The main hazards were drilling, cutting, water spills and pinch points in the linkage. We controlled these risks by wearing PPE, clamping timber before drilling, keeping fingers clear of moving arms and cleaning water leaks before testing.

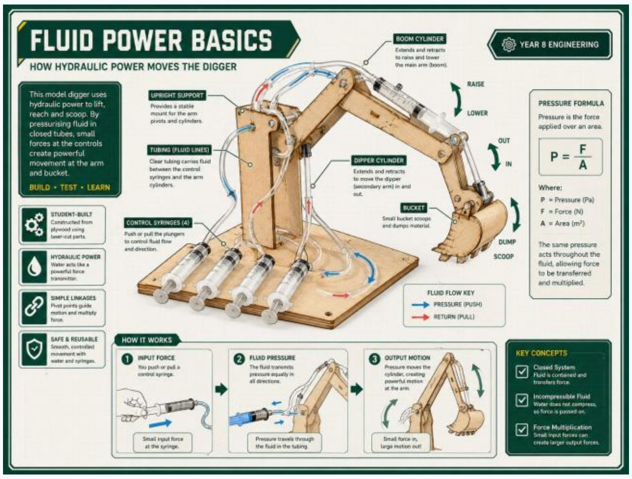
BEFORE YOU TICK DONE

- ☐ Hazards named
- ☐ Controls explained
- ☐ Testing safe

BEST PRACTICE ANSWER

FLUID POWER BASICS

A strong answer uses Pascals principle to explain how force is transferred through trapped liquid.



1 Pascals principle

- Force on one piston creates pressure
- Pressure transfers through trapped water
- Output piston moves the linkage

2 Pressure formula

- Pressure = Force / Area
- Larger piston area changes output force
- Use correct units in calculations

3 Control problems

- Air makes movement spongy
- Leaks reduce control
- Kinks stop smooth flow

OK

BEST PRACTICE STUDENT RESPONSE

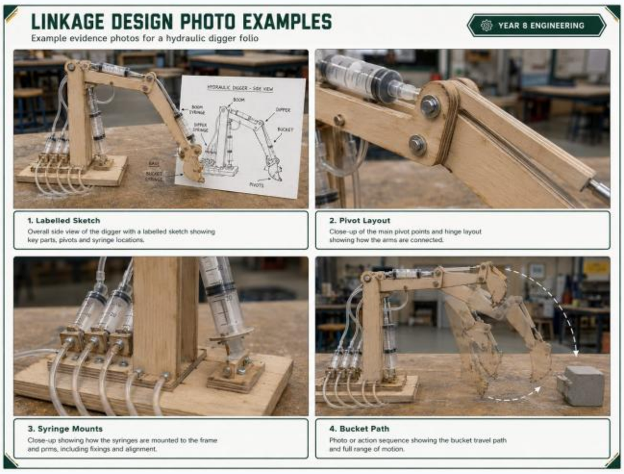
Pascals principle means pressure applied to trapped water is transferred through the tubing. The output syringe moves because pressure acts on its piston area. Air bubbles or leaks reduce control, so the system must be filled and sealed carefully.

BEFORE YOU TICK DONE

- ☐ Pascal explained
- ☐ Formula used
- ☐ Air/leaks linked

LINKAGE DESIGN

A strong answer documents the arm geometry, pivot points, syringe mounts and expected movement path.



1 Labelled sketch

- Show boom, dipper and bucket
- Label fulcrum, load and effort points
- Mark syringe mount positions

2 Movement path

- Predict bucket curl path
- Check for binding or collision
- Show expected travel direction

3 Photo evidence

- Labelled Sketch
- Pivot Layout
- Syringe Mounts
- Bucket Path

OK

BEST PRACTICE STUDENT RESPONSE

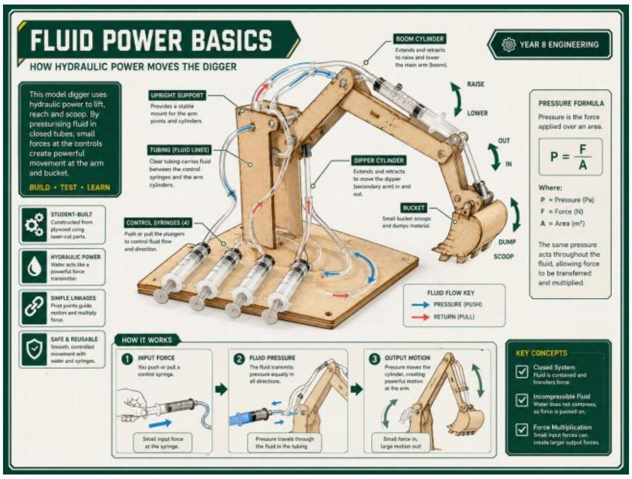
Our linkage sketch shows the boom, dipper, bucket, pivot points and syringe mounts. We placed the syringe mounts to give controlled movement without binding. The bucket path sketch shows how the bucket should curl and where movement may be limited.

BEFORE YOU TICK DONE

- ☐ Pivots labelled
- ☐ Mounts shown
- ☐ Path explained

SYRINGE SIZING

A strong answer records syringe choices and explains the force-travel trade-off using piston area.



1 Syringe choice

- Record input and output syringe sizes
- Match sizes to boom, dipper and bucket
- Explain why each pair was chosen

2 Calculations

- Find piston area
- Use pressure and force ideas
- Show units clearly

3 Trade-off

- Larger output gives more force
- More force usually means less travel
- Compare prediction with testing

OK

BEST PRACTICE STUDENT RESPONSE

We chose syringe sizes by thinking about force and travel. A larger output syringe can create more lifting force, but it moves a shorter distance. Our calculations used piston area to predict how the boom, dipper and bucket would behave.

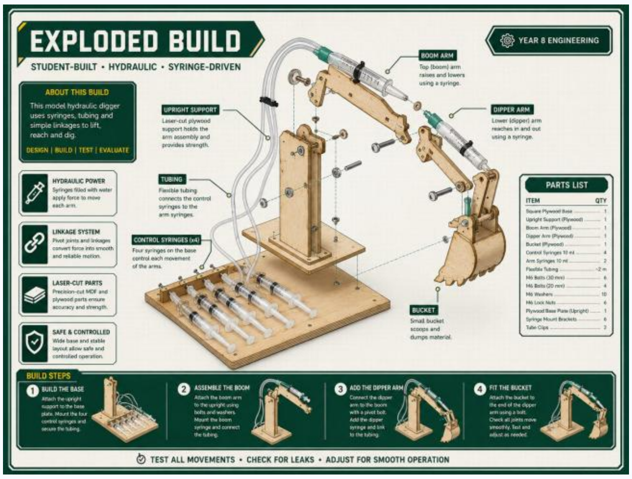
BEFORE YOU TICK DONE

- ☐ Sizes recorded
- ☐ Areas calculated
- ☐ Trade-off explained

BEST PRACTICE ANSWER

BUILD PLAN

A strong answer shows the planned construction sequence before parts are cut and assembled.



1 Sequence

- Build base first
- Add boom, dipper and bucket
- Install pivots before hydraulics

2 Accuracy checks

- Mark and drill holes accurately
- Keep arms stiff and aligned
- Check movement before final fixing

3 Hydraulic layout

- Plan tidy hose routing
- Avoid kinks and sharp bends
- Leave room for syringe travel

OK

BEST PRACTICE STUDENT RESPONSE

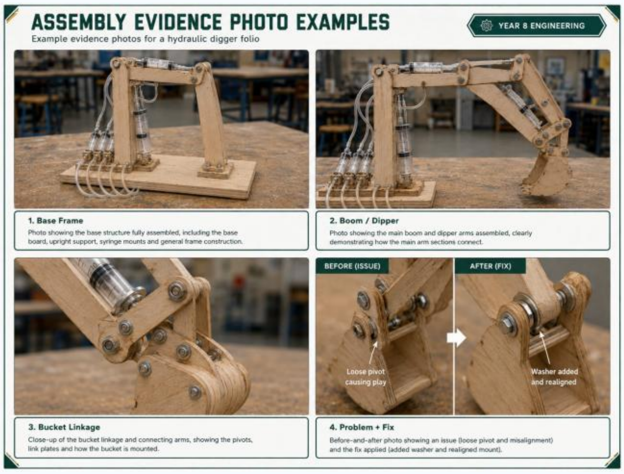
Our build sequence was base, boom, dipper, bucket, pivots and then hydraulic circuits. We checked accuracy by marking hole positions carefully and testing arm movement before final assembly. Tidy hose routing was needed to avoid kinks and leaks.

BEFORE YOU TICK DONE

- ☐ Sequence clear
- ☐ Accuracy checked
- ☐ Hoses planned

ASSEMBLY EVIDENCE

A strong answer uses build-log evidence to explain decisions, problems, fixes and quality checks.



1 Build photos

- Base Frame
- Boom / Dipper
- Bucket Linkage
- Syringe Mounts

2 Problems and fixes

- Name one issue clearly
- Explain what caused it
- Show the fix with evidence

3 Quality checks

- Pivots move freely
- Arms are stiff and aligned
- Mounts are secure

OK

BEST PRACTICE STUDENT RESPONSE

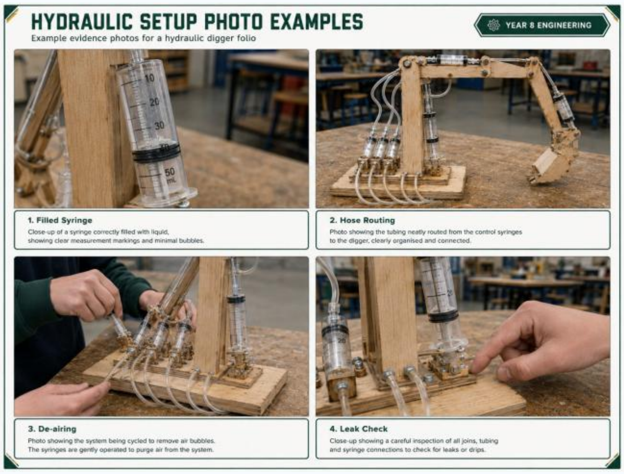
During assembly we built the base, boom, dipper and bucket linkage, then checked that each pivot moved freely. One problem was a loose pivot, which caused poor bucket control. We fixed it by adding a washer and realigning the mount.

BEFORE YOU TICK DONE

- ☐ Photos included
- ☐ Problem fixed
- ☐ Quality checked

HYDRAULIC SETUP

A strong answer documents filling, plumbing, de-airing, leak checks, hose routing and syringe mounting.



1 Fill and connect

- Fill syringes with water
- Connect tubing securely
- Route hoses to avoid kinks

2 De-air system

- Remove bubbles from tubing
- Cycle syringes before testing
- Refill if movement feels spongy

3 Leak check

- Filled Syringe
- Hose Routing
- De-airing
- Leak Check

OK

BEST PRACTICE STUDENT RESPONSE

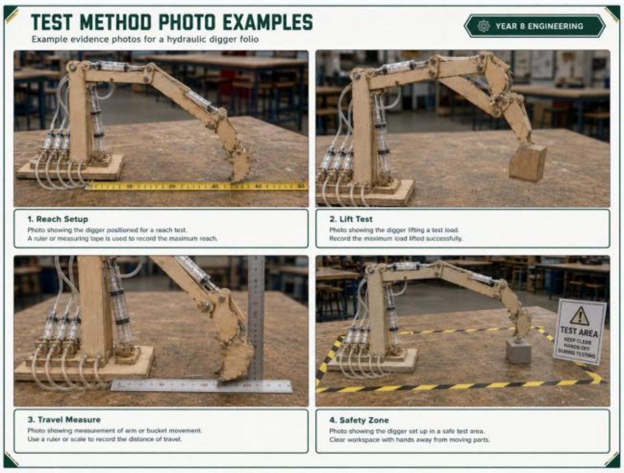
We filled each hydraulic circuit with water and connected the tubing securely. We removed air by cycling the syringes until movement was smooth. The system worked best when there were no bubbles, no kinks and no leaks at the joins.

BEFORE YOU TICK DONE

- ☐ Filled shown
- ☐ Air removed
- ☐ Leaks checked

TEST METHOD

A strong answer describes a fair, repeatable setup another group could follow safely.



1 Fair setup

- Use the same start position
- Keep the base in the same place
- Use the same load for lift tests

2 Measurements

- Measure reach
- Measure lift or load capacity
- Measure input/output travel

3 Safety zone

- Clear the test area
- Keep hands away from linkages
- Record results straight away

OK

BEST PRACTICE STUDENT RESPONSE

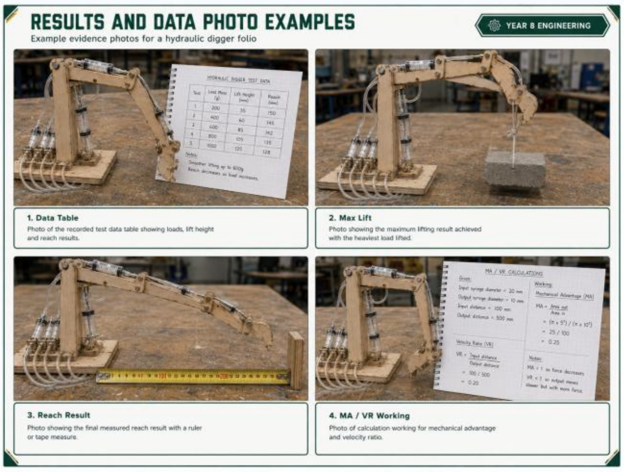
To test the digger, we used the same start position each time and measured reach, lift and travel with a ruler. The test was fair because the base position and load were kept the same. We kept a clear safety zone during movement.

BEFORE YOU TICK DONE

- ☐ Fair setup
- ☐ Measures named
- ☐ Safety clear

RESULTS AND DATA

A strong answer reports measured performance and interprets reach, lift, movement, MA, VR and efficiency.



1 Data table

- Record reach and lift results
- Record input and output travel
- Use clear headings and units

2 Performance

- Maximum lift
- Reach result
- Smooth movement observations

3 Calculations

- Mechanical advantage
- Velocity ratio
- Efficiency or losses

OK

BEST PRACTICE STUDENT RESPONSE

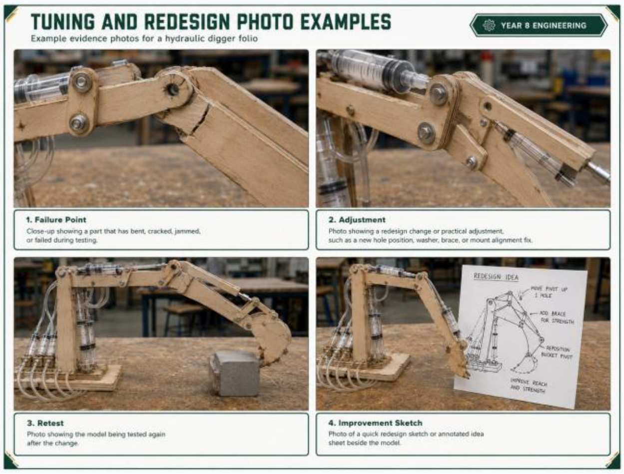
The results table shows our measured reach, lift and input/output travel. The maximum lift was recorded during a controlled test. The MA and VR results show the trade-off between force and movement, with losses caused by friction, flex and small leaks.

BEFORE YOU TICK DONE

- ☐ Data table
- ☐ Units used
- ☐ Results interpreted

TUNING AND REDESIGN

A strong answer analyses what limited performance and proposes a specific improvement.



1 Find the limit

- Leaks, air or friction
- Flex in arms or mounts
- Binding at pivots

2 Make one change

- Adjust mount position
- Tighten or align a pivot
- Improve hose route or seal

3 Retest and explain

- Failure Point
- Adjustment
- Retest
- Improvement Sketch

OK

BEST PRACTICE STUDENT RESPONSE

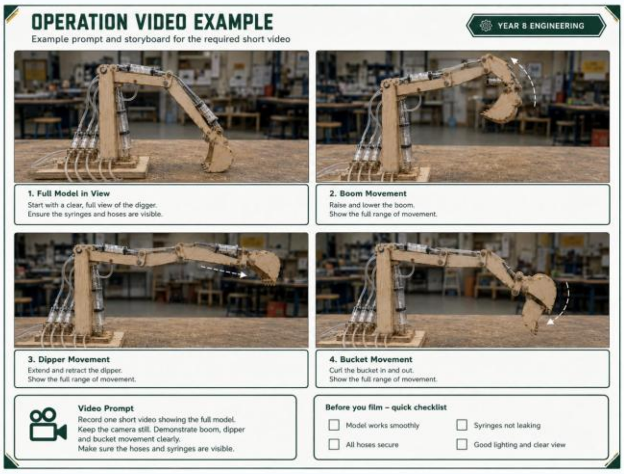
The first limit was flex in the arm and friction at one pivot. This reduced smooth movement and lifting control. We improved it by realigning the pivot and would strengthen the arm next time. The retest showed smoother movement but still some loss from friction.

BEFORE YOU TICK DONE

- ☐ Limit named
- ☐ Change shown
- ☐ Trade-off explained

FINAL EVALUATION

A strong answer makes a final judgement that links evidence, engineering theory, WHS and improvements.



1 Judge success

- What worked best
- What failed first
- Use evidence, not guesses

2 Link theory

- Hydraulic pressure transfer
- Force-travel trade-off
- Friction, leaks and air loss

3 Reflect and improve

- Name a practical improvement
- Explain why it would help
- Refer to operation video evidence

OK

BEST PRACTICE STUDENT RESPONSE

Overall, our digger met the brief because it produced controlled boom, dipper and bucket movement. The strongest evidence is the test data, photos and operation video. This project taught me that small leaks, air bubbles and pivot friction can strongly affect hydraulic performance.

BEFORE YOU TICK DONE

- ☐ Judgement made
- ☐ Evidence linked
- ☐ Improvement named